

CSE107 Lab 6 Report

Lab Date: 07/11/2025 — Due: next lab day

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Section: Section 2
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This report has 3 sections. Keep it short, clear, and readable. Submission format: a single PDF (export this file to PDF and upload).

1) Lab Content & What We Did (Short Summary)

The main objective of this labwork was to strip HTML tags from a text using the C programming language, leaving only the plain text content. For this purpose, I read the input file named labwork6.html and wrote the tag-free text to the output file named out.txt. In the implementation, I used a flag to determine if a character was inside an HTML tag, starting with < and ending with >. The program wrote the character if the flag was 0 and skipped it if the flag was 1.

2) Function Description

- fopen: fopen is used to open a file so the program can use it. It takes the file's name and the access mode as input. Its main purpose is to give you a special FILE* pointer, which acts like a handle or key, allowing all other file functions to interact with that specific file.
- fclose: fclose is essential for closing a file that you opened earlier using FILE*. When you close a file, the program saves any unsaved data still waiting in memory (the buffer) to the disk. It also properly releases the computer's resources, like memory, that the file was using.
- fgetc: fgetc stands for "file get character." This function reads just one character at a time from the file. We typically use it inside a loop to go through the entire file's content, character by character. When the reading process hits the end of the file, it returns the special signal EOF (End Of File).

- fputc: fputc stands for "file put character." This function writes a single character to the specified output file. In the labwork, I used it to take the clean, non-tag characters from the input file and place them into the output file, one character at a time.

- Explain the use of FILE*. What is it used for?

The FILE* is a special pointer used by C to manage files. It is like the file's ID badge once it's opened. This badge holds all the important information about the file, such as where it is on the disk and what your current position is inside it (which character you are about to read or write next). All file functions, like fgetc and fputc, must use this FILE* to know exactly which file to work on.

3) Your Code

```
#include <stdio.h>

#include <stdlib.h>

int main()
{
    int c;

    int flag = 0; // state variable: 0 = plain text, 1 = inside a tag

    FILE *fp = fopen("labwork6.html", "r"); // opens the HTML file in read mode

    if(!fp)
    {
        perror("failed to open labwork6.html.\n");

        return (1);
    }

    FILE *out = fopen("out.txt", "w+"); // opens the output file in write mode

    if(!out)
    {
        perror("failed to open out.txt.\n");

        fclose(fp);

        return (1);
    }
}
```

```
while((c = fgetc(fp)) != EOF)
{
    if(c == '<')
        flag = 1; // set flag 1, do not print the character yet

    if(flag == 0)
        fputc(c, out); // if flag is 0 write the character to the output file

    if(c == '>' && flag == 1)
        flag = 0;
}

fclose(fp);

fclose(out);

return(0); // success
}
```